

**GOVINDRAM SEKSARIA INSTITUTE OF
TECHNOLOGY & SCIENCE, INDORE
END SEMESTER EXAMINATION**

B.E. — First Year (Mechanical Engineering)
Maximum Marks: 56 Time: 3 Hours

SET F

UNIT I — Mechanical Properties (14 Marks)

- Q1.** (a) Define modulus of rigidity. (2 Marks) (b) Define necking. (2 Marks)
Q2. Explain true stress and true strain. (3 Marks)
Q3. Describe creep mechanism and factors affecting creep. (7 Marks)

UNIT II — Theory of Machines (14 Marks)

- Q4.** (a) Define follower displacement. (2 Marks) (b) Define dwell period. (2 Marks)
Q5. Differentiate between uniform motion cam and SHM cam. (3 Marks)
Q6. Explain velocity analysis of a four-bar mechanism using graphical method. (7 Marks)

UNIT III — Fluid Kinematics (14 Marks)

- Q7.** (a) Define Reynolds number. (2 Marks) (b) Define stagnation point. (2 Marks)
Q8. Explain classification of flows based on Mach number. (3 Marks)
Q9. Derive expression for volumetric flow rate using velocity field. (7 Marks)

UNIT IV — IC Engines & Automobiles (14 Marks)

- Q10.** (a) Define dead centre. (2 Marks) (b) Define piston speed. (2 Marks)
Q11. Explain supercharging and its advantages. (3 Marks)
Q12. Explain working of turbocharger with neat diagram. (7 Marks)